Shukla Ashish Anand

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Education

Indian Institute of Science Education and Research (IISER), Bhopal — Bhopal, MP

Bachelor of Technology (B.Tech) in Data Science and Engineering, ***CPI: 8.22 / 10.0*** — *Aug. 2024 – May 2028 (Expected)*

- **Relevant Coursework**: Machine Learning, Computer Vision, Optimization Techniques, Probability & Random Processes, Signals & Systems, Data Structures & Algorithms, Linear Algebra, Computer Organisation, Discrete Mathematics.

Technical Skills

Research Interests: Robust & Efficient ML, Multimodal Learning, Speech & Audio Processing, Representation Learning, Domain Adaptation, Test-Time Adaptation.
Languages: Python, C/C++, SQL.
ML & Deep Learning: PyTorch, Hugging Face Transformers, PEFT, LoRA, DoRA, QLoRA, bitsandbytes.
Speech & Multimodal: Whisper, Qwen3-ASR, LAION-CLAP, torchaudio, librosa, scipy.signal, AVSR, VideoMAE, Mamba-2.
Tools & Systems: Git, Linux/Bash, Weights & Biases, Remote GPU Clusters, NumPy, Pandas, OpenCV, SQLite (FTS5).

Experience

Speech Lab, CCDS @ Nanyang Technological University (NTU) — Remote

Research Intern – Speech Data & ASR Evaluation — *Aug. 2026 – Dec. 2026*

- Working on **speech data quality, accent robustness, and Automatic Speech Recognition (ASR)**, studying how data curation and language identification affect the reliability and representativeness of speech datasets.
- Evaluating **Whisper and Qwen3-ASR** on accented English, investigating the effects of data filtering, text normalization, decoding strategies, and parameter-efficient adaptation on recognition performance.

Speech Lab, LTRC @ IIIT Hyderabad — Hyderabad, India

Research Intern – Speech & Audio Modeling — *May 2026 – Jul. 2026*

- Developed an information-theoretic pitch stylization pipeline using Shannon Entropy, Mutual Information, and KL Divergence for robust speech representation modeling.
- Applied Dynamic Programming and Weighted Least Squares regression for optimal piecewise pitch contour segmentation.
- Derived a Bayesian Information Criterion (BIC/MDL) for automatic model-complexity selection, achieving a **38% RMSE reduction** over heuristic baselines.

Visual Data Computing Group (VisDom) @ IISER Bhopal — Bhopal, India

Undergraduate Researcher – Multimodal Machine Learning — *Jan. 2025 – Present*

- Working under Dr. Vinod K. Kurmi on robust multimodal and speech modeling (**PRISM, Mamba-AVSR, and VaaniAdapt**), focusing on test-time adaptation under distribution shifts and linear-time multimodal architectures.

Publications & Submissions

PRISM (First Author) | *Published in ACM CIKM 2026 (Full Paper, CORE A)* | DOI | Code | arXiv — Nov. 2026

- **Prototype-Rectified Iterative Self-supervised Manifold Denoising under Severe Acoustic Shift** (pp. 4349–4359).
- Conceived and developed a **training-free, source-free test-time adaptation (TTA)** framework for audio-text foundation models under severe acoustic domain shifts.
- Modeled acoustic noise as a low-rank affine distortion in multimodal latent space and reversed it via closed-form geometric corrections using **frozen text prototypes as anchors**, reducing adaptation to a single matrix-vector multiplication in **0.0009 ms**.
- Improved over the zero-shot baseline by **+12.94 pp** and surpassed an oracle-assisted TTA baseline by **+9.41 pp** on UrbanSound8K, while resolving broadband subspace deflation via Confidence-Aware Regression (**+8.16 pp** on worst-affected classes).

Pitch Stylization | *Under Review at IEEE ICASSP 2027* — 2026

- **Mutual-Information-Guided Multi-Estimator Pitch Fusion for Robust Pitch Stylization.**
- Developed an information-theoretic multi-estimator fusion framework combining complementary pitch estimators with dynamic programming for prosodic contour stylization.

Research Projects

Mamba-AVSR | *PyTorch, Mamba-2, Whisper-v3, VideoMAE, Llama-3.2* Dec. 2025 – Present

- Prototyped an end-to-end Audio-Visual Speech Recognition pipeline integrating Whisper-v3 and VideoMAE encoders with a Llama-3.2 backend.
- Replaced quadratic $O(N^2)$ cross-attention with a linear-time $O(N)$ Mamba-2 connector to efficiently fuse asynchronous audio and video streams under acoustic corruption.

VaaniAdapt | *Continual Learning, Multilingual ASR* 2026

- Designed a rehearsal-free continual learning approach using orthogonality-constrained low-rank residual adapters to adapt deployed ASR models to new languages without catastrophic forgetting or prior data replay.

DermaScribe | *System Architect & AI Lead, Python, SQLite (FTS5), OpenCV* Nov. 2025

- Architected a hybrid offline-online clinical compatibility database using SQLite and FTS5 indexing with an OpenCV screening module, achieving sub-millisecond query latency.

Academic Grants & Recognition

- **ACM SIGIR Student Travel Grant (2026)**: Recipient of the ACM SIGIR student travel grant supporting the first-author full paper (*PRISM*) at ACM CIKM 2026.
- **IISER Bhopal DSE Day Hackathon (2025)**: Placed 1st in the Healthcare Track for architecting *DermaScribe*.